MIL-P-52457B(ME)
15 July 1977
SUPERSEDING
MIL-P-52457A(ME)
11 July 1973

MILITARY SPECIFICATION

PANEL, DISTRIBUTION, BATTERY CHARGING

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a military design distribution panel used for charging lead-acid type batteries when connected to a 28 volt DC power source.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 6130

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized and Containerized Unit Loads, 40" X 48" Pallets, Skids, Runners, or Pallet Type Base.
MIL-STD-810	- Environmental Test Methods.

DRAWINGS

ME

TA13208E4850	- Panel, Distribution, Battery Charging.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The panel, distribution, battery charging (hereinafter referred to as "panel") shall be in accordance with TA13208E4850 and as specified herein.

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the panels. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.1.2 Design and operation. The panel contains two manually controllable, 8 amps current limited charging stations for 12 V batteries, two manually controllable 8 amps current limited charging stations for 24 V batteries, and eight noncontrollable 20 amps current limited charging stations for 24 V batteries. In addition, the panel may be used to charge series arrangements of 6 V batteries or 12 V batteries when connected to the proper station. Each station is independently current limited and protected by circuit breakers. For each manually controlled charging station, the charging terminal voltage and charging current are manually controlled by variable resistors and the magnitudes are determined by an internal voltmeter-ammeter arrangement that is inserted into the proper circuit through a selector switch. The terminal voltage of discharged batteries may be determined by connecting to an unused controlled charging station with the circuit breaker open and setting the selector switch to that station. For each uncontrolled charging station, the charging voltage is equal to the terminal voltage of the power source and no voltage or current measurement circuits are provided.

3.1.3 Conflicting requirements. In the event of any conflict between the requirements of this specification and the referenced documents, the requirements of this specification shall take precedence.

3.2 First article (preproduction model). The contractor shall furnish one or more panels as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce panels that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

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3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more panels for testing as specified in 4.5. The panels furnished shall be selected from those panels that have passed the individual tests required by quality conformance inspection.

3.4 Functional characteristics. There shall be no evidence of incorrect wiring, incorrect instrument indications, damage to wiring or components or failure to replace any component that does not operate within its specified characteristics when each station is tested as specified in 4.6.1.

3.5 Reverse current. The reverse current through the circuit of any station shall not exceed 0.25 milliamp at an applied voltage of 50 volts DC after each station has been tested and deficiencies corrected as specified in 4.6.2.

3.6 Operation. There shall be no erratic operation, component failure or wiring damage, when all or any designated number of stations are operated simultaneously at any current up to and including the values limited by the trip characteristics of the circuit breakers. This test shall be performed as specified in 4.6.3.

3.7 Vibration. There shall be no breakage, deformation, or degradation of any part, component or electrical connection that requires maintenance action prior to or during performance of the functional characteristics test specified in 4.6.1 after the panel has been vibration tested as specified in 4.6.4.

3.8 Shock (specified pulse). There shall be no breakage, deformation or degradation of any part, component or electrical connection that requires maintenance action prior to or during performance of the functional characteristics test specified in 4.6.1 after the panel has been shock tested as specified in 4.6.5.

3.9 Temperature-humidity. There shall be no evidence of corrosion formation, damage resulting from moisture accumulation or deformation of material that would result in failure of any station of the panel to perform its intended function after the panel has been temperature-humidity cycled as specified in 4.6.6.

3.10 Low temperature storage. There shall be no breakage, deformation or loosening of any part, component, or electrical connection that would result in failure of any station to perform its intended function after the panel has been temperature stabilized at minus 54° C (-65° F), maintained at this temperature for a 24 hour period, and the temperature adjusted to rise and stabilize at minus 32° C (-25° F) prior to performance of the low temperature operation test. The procedure shall be as specified in 4.6.7.

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3.10.1 Low temperature operation. There shall be no breakage, deformation or degradation of any part, component or electrical connection that requires maintenance action prior to or during performance of the functional characteristics test specified in 4.6.1 when this test is performed in an ambient temperature of minus 32° C (-25° F) and the panel shall operate for 30 minutes without failure in this same ambient environment. This test shall be performed as specified in 4.6.8.

3.11 High temperature storage. There shall be no breakage, deformation or loosening of any part, component, or electrical connection that would result in failure of any station to perform its intended function after the panel has been temperature stabilized at 65° C (149° F), maintained at this temperature for a 24 hour period, and the temperature adjusted to lower, and stabilize at 50° C (122° F) prior to performance of the high temperature operation test. The procedure shall be as specified in 4.6.9.

3.11.1 High temperature operation. Each station of the panel shall operate continuously for 4 hours without failure at a temperature of 50° C (122° F) when the panel is tested as specified in 4.6.10.

3.12 Reliability evaluation (high temperature). The reliability shall be evaluated by the Government as part of initial production tests. The panels shall be operated as specified in 4.6.3, except in an ambient temperature range of 40° to 50° C (104° to 122° F) and the loads transferred on the uncontrolled stations only as permitted in the test procedure. Testing shall be performed as specified in 4.6.11.

3.13 Maintainability. During performance of reliability test, the time required to perform each maintenance action when a failure occurs shall not exceed 4 manhours for each 100 hours of operation. This time shall include all functions necessary for repair and checkout by qualified maintenance personnel after all preparations for the maintenance action has been made. Spare parts to support this requirement shall be as specified (see 6.2).

3.14 Safety. The panel shall be properly grounded with its existing ground terminal, before any test or operation, as a general safety precaution.

3.15 Workmanship. Panels shall be examined for cross threaded fasteners, misalined and improperly set rivets, incomplete fillet welds, reduced diameter spot welds, cold or weak solder joints, smudged or illegible lettering, errors in assembly and other manufacturing defects that could affect end item operation and maintenance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article preproduction (see 4.3).
(b) Quality conformance (see 4.4).
(c) Initial production (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The components and assembled panel shall be examined as specified.

4.3.1.1 Components. Each component shall be examined for compliance with applicable drawings, specifications, and standards.

4.3.1.2 Assembled panel. The assembled panel shall be examined for compliance with the applicable assembly drawings, wiring, and workmanship.

4.3.2 Tests. The preproduction panels that have passed examination requirements shall be subject to the tests marked "X" in Column 1 of the Test Schedule and in the sequence listed.

4.3.3 Inspection failures. Failures occurring during preproduction examination shall be corrected or resolved prior to initiation of testing. If failure occurs during any test, testing shall be discontinued and the Government representative responsible for inspection shall be directly notified. Unless otherwise instructed by the local Government representative, the failure areas shall be isolated and corrected and testing resumed until the remainder of tests specified in the Test Schedule have been completed. The type of failure, a determination as to cause and the related facts, including corrective action taken, shall be included in the test report. Failure to meet the specified requirements shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Examination. Examination shall be performed on samples of fully assembled panels. Samples shall be selected in accordance with MIL-STD-105 for special Inspection Level S-4. The AQL shall be 2.5 percent defective. Examination shall consist of verifying that all parts, components, assemblies, and workmanship are in compliance with the specified requirements.

4.4.2 Tests - individual and samples.

4.4.2.1 Individual. Each panel shall be subject to the tests marked "X" in Column 2 of the Test Schedule.

4.4.2.2 Samples. Panels used for sample tests shall be selected from those that have passed the examination requirements. The number selected shall conform with Inspection Level S-2 of MIL-STD-105 and the AQL shall be 6.5 percent defective. Selected samples shall be subject to the tests marked "X" in Column 3 of the Test Schedule.

4.4.3 Inspection failure. Failure to meet the specified requirements for quality conformance inspection shall be cause for rejection.

4.5 Initial production inspection. When specified (see 3.3), one or more initial production panels will be selected at random by the Government from the panels being produced by production tooling and will be inspected as specified in 4.5.1 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of the initial production panels shall not exclude the remaining panels from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any tests specified as part of the initial production tests, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production tests and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.5.1 Inspection of initial production models.

4.5.1.1 Examination. The assembled panels shall be examined for missing components, loose, or improperly mated parts and workmanship.

4.5.1.2 Tests. For initial production testing, the panels shall be divided into two groups (see 6.2). The panels selected for group 1 tests shall be subjected to the tests marked "X" in Column 4 of the Test Schedule, except reliability tests. The panels selected for group 2 tests shall be subjected to only the reliability test of the Test Schedule. Both groups of panels should be scheduled for concurrent testing.

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4.5.1.3 Special requirement. The panels selected for group 1 tests shall use a set of batteries for performance of the operation test specified in 3.6 and be operated only for the length of time necessary to verify that performance is satisfactory.

4.5.2 Inspection failure. Failure of an initial production panel to meet any requirements specified herein during and as a result of the inspection specified in 4.5 shall be cause for rejection of the initial production panels and shall be cause for refusal by the Government to continue acceptance of production panels until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on panels previously produced and accepted under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all panels accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.6 Inspection procedures. Unless otherwise specified, all tests shall be performed at an ambient temperature of 10° to 32° C (50° to 90° F). For tests specified to be performed outside this temperature range, the contractor shall assure that all test equipment is properly rated. Tolerances on specified temperatures shall be ± 1.0° C and tolerances on voltages, currents, and related electrical quantities shall be 3 percent. The power rating for designated loads is for continuous operation, however, ratings that meet testing duty cycle requirements may be used.

Test Schedule

First Article	Qual Conf Ind	Qual Conf Sample	Init Prod Insp Comp	Test	References Req	References Test
1	2	3	4	5	6	7
X	X	-	X	Functional characteristics.	3.4	4.6.1
X	X	-	X 1/	Reverse current.	3.5	4.6.2
X	-	X	X 1/	Operation.	3.6	4.6.3
X	-	-	X	Vibration.	3.7	4.6.4
X	-	-	X	Shock.	3.8	4.6.5
X	-	-	-	Humidity.	3.9	4.6.6
X	-	-	-	Low temperature storage.	3.10	4.6.7
X	-	-	X	Low temperature operation.	3.10.1	4.6.8
X	-	-	-	High temperature storage.	3.11	4.6.9
X	-	-	X 2/	High temperature operation.	3.11.1	4.6.10
-	-	-	X 2/	Reliability evaluation.	3.12	4.6.11

1/ See 4.5.1.3 - Special requirement for Group 1 tests.
2/ See 4.5.1.2 - Group 2 Testing.

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4.6.1 Functional characteristics. Each station shall be tested individually, as specified, with the designated load. All designated loads may be preconnected or they may be transferred as required to the specified stations. For all tests on uncontrolled stations, an ammeter shall be inserted into the test circuit. Establish the pretest conditions as follows: Set all switches and all circuit breakers to the off position. Set the knobs of the controlled stations completely counterclockwise. Connect a 28 volt DC power source to the panel power connector. Close all power switches so that voltage is present at the input side of the circuit breaker. Connect the designated load to the station to be tested.

(a) Controlled 12 V stations (1 and 4).

Designated load. Load these stations with a nominal value resistor of 1.5 ohms having a minimum power rating of 300 watts.

Test procedure. Set the instrument selector switch to the station to be tested and close the circuit breaker. The initial instrument readings should be about 11.2 volts and 5.6 amps. Increase the panel control knob setting until the ammeter indicates about 8 amps. The voltmeter should read about 16 volts. After the circuit breaker has held for 15 seconds minimum, increase the control knob setting until the ammeter indicates 12 amps. The voltmeter should indicate about 24 volts and the circuit breaker should trip in the range from 3 to 25 seconds. If breaker fails to trip in less than 25 seconds, remove power to the station, correct the defects and retest.

(b) Controlled 24 V stations (2 and 3).

Designated load. Load these stations with a nominal value resistor of 2.3 ohms having a minimum power rating of 350 watts.

Test procedure. Set the instrument selector switch to the station to be tested and close the circuit breaker. The initial instrument readings should be about 14.5 volts and 6.2 amps. Increase the control knob setting until the ammeter indicates about 8 amps. The voltmeter should read about 18.6 volts. After the circuit breaker has held for 15 seconds minimum, increase the control knob setting until the ammeter indicates 12 amps. The voltmeter should indicate about 28 volts and the breaker should trip in the range from 3 to 25 seconds. If breaker fails to trip in less than 25 seconds, remove power to the station, correct the defects and retest.

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(c) Uncontrolled 24 V stations.

Designated load. Load these stations with a nominal value resistor of 1.5 ohms having a minimum power rating of 525 watts and a nominal value 3.0 ohms resistor having a minimum power rating of 300 watts that can be connected to the 1.5 ohms resistor through a switch to form a parallel load.

Test procedure. With an ammeter in the circuit and the initial load set for 1.5 ohms, close the circuit breaker to the station under test. The current should be about 18.7 amps. After the breaker holds for a minimum 15 seconds, switch the 3.0 ohm resistor in parallel and observe that the current increases to about 28 amps. If the breaker fails to trip in the range from 3 to 30 seconds, remove power to the station, correct the defects and retest.

Failure to correct any defect and satisfactorily complete all tests shall be cause for rejection.

4.6.2 Reverse current. Set all circuit breakers and the instrument selector switch to the off positions. The resistors of controlled stations may be set to any position but minimum resistance is preferred. Short circuit the power input connector (J13) and the A1-A2 contacts of the power relay. Select a power source containing integral measuring, control and protective circuits or devise an arrangement that can perform these functions. Establish a voltage of 50 volts DC minimum across the terminals of one of the panel charging receptacles, close the circuit breaker, measure and record the reverse current and reopen the breaker. Measure and record the reverse current for each station. If the reverse current exceeds 0.25 milliamp for any station, correct the defects and retest. Failure to correct any defects shall be cause for rejection.

4.6.3 Operation. The panel shall be tested with the designated loads and in accordance with the specified test procedure.

(a) Designated loads.

Controlled 12 V stations (1 and 4). Load these stations with a nominal value 1.0 ohm resistor having a minimum power rating of 50 watts.

Controlled 24 V stations (2 and 3). Load these stations with a nominal value 1.5 ohm resistor having a minimum power rating of 100 watts.

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Uncontrolled 24 V stations. Load these stations with a nominal of 1.5 ohm resistor having a minimum power rating of 600 watts.

(b) Test procedure. Set the knobs of the controlled stations completely counterclockwise so maximum resistance is in the circuit. Set all panel switches and circuit breakers to off. Connect a 28 volt DC power source having a minimum current capacity of 105 amps to the panel power receptacle (J13). Connect a designated load to one of the uncontrolled station through an ammeter and operate the setup to verify that the load is set for a current flow of about 18.7 amps. Repeat for three other uncontrolled stations and connect the loads to each of these stations. Connect the designated loads to each of the controlled stations and verify that the current flow to each station is about 7 amps by use of the panel instruments. After preliminary checkout, operate the panel for 4 hours with the eight stations loaded. After 4 hours, disconnect the loads from the four uncontrolled stations and reconnect them to the other four that were previously unused. Continue to operate the panel for an additional 4 hours using eight stations. Nonconformance to the specified requirements shall constitute failure of this test.

4.6.4 Vibration. Test the panel, complete with cable assemblies stowed and case halves secured together as specified in Method 514.2 of MIL-STD-810. The test shall be performed in accordance with Procedure X, Figure 514.2-VII and curve AW as required for equipment transported as secured cargo. Test duration shall be 84 minutes for each of three mutually perpendicular axes. Following the completion of testing, make a damage assessment and perform the functional characteristics test specified in 4.6.1. Any damage that requires maintenance action, component replacement or results in performance outside of the specified limits shall constitute failure of this test.

4.6.5 Shock (specified pulse). Test the panel, complete with cable assemblies stowed and case halves secured together, in accordance with MIL-STD-810C, Procedure III except as specified herein. The pulse characteristics shall be half sine, 60g and 11 milliseconds. Impact each of four surfaces (top, bottom, front side and back side) with two consecutive impulses each for a total of eight impacts. For front side and back side tests, secure the panel to the shock platform using plywood spacers to provide clearance for the case latches. Do not test on case ends. Make a damage assessment following each pair of impulses. At the completion of shock tests perform the functional characteristics

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test specified in 4.6.1. Any damage that requires maintenance action, component replacement, or results in performance outside of the specified limits shall constitute failure of this test.

4.6.6 Humidity. Subject the panel, complete with cable assemblies stowed and case halves open, to the 48 hour humidity cycle as shown on Figure 1. At the completion of the cycle, remove from humidity environment and allow the panel to dry for 24 hours in the local ambient environment. Examine for damage. Any visible damage or subsequent operational defects traceable to humidity exposure shall constitute failure of this test.

4.6.7 Low temperature storage. Subject the panel, complete with cable assemblies stowed and case halves open to a temperature of minus 54° C. The complete assembly shall be stabilized and then maintained at this temperature for 24 hours. After the 24 hour period, raise the temperature to minus 32° C, allow the panel to stabilize at this temperature, make a damage assessment and proceed to the low temperature operation test. Any damage that requires corrective action prior to initiation of low temperature performance testing shall constitute failure of this test.

4.6.8 Low temperature operation. Following the successful completion of low temperature storage and with panel maintained in the ambient environment of minus 32° C, test the panel for the functional characteristic as specified in 4.6.1. After functional characteristics checkout, connect a resistive load to each station using the cable assemblies tested with the panel. The load on each controlled station shall produce a minimum current flow of 5 amps when the panel controls are set at midrange and the load for each uncontrolled station shall produce a minimum current flow of 10 amps. Operate the panel with these loads for 30 minutes. Any damage that requires corrective action during performance of this test shall constitute failure of this test.

4.6.9 High temperature storage. Subject the panel, complete with cable assemblies stowed and case halves opened to a temperature of 65° C. The complete assembly shall be stabilized and then maintained at this temperature for 24 hours. After the 24 hour period, lower the temperature to 50° C, allow the panel to stabilize at this temperature and proceed to the high temperature operation test. Any damage that requires corrective action prior to initiation of high temperature operation test shall constitute failure of this test.

4.6.10 High temperature operation. Following the successful completion of high temperature storage and with the panel maintained at 50° C, connect a resistive load to each station using the cable assemblies tested with the panel. The load on each controlled station shall produce a minimum current flow of 5 amps when the panel controls are set at midrange and the load for

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each uncontrolled station shall produce a minimum current flow of 10 amps. Operate the panel with these loads for 2 hours. Nonconformance to the specified requirements shall constitute failure of this test.

4.6.11 Reliability evaluation (high temperature). The initial production panels shall be tested under the conditions specified in 3.12. The accumulated hours of testing shall be equal to 300 hours times the number of panels used for the test. If failure occurs on any uncontrolled station, the failure shall be recorded in operational hours to failure, the load shall be transferred to an unused station and testing continued. If failure occurs on a controlled station or in another area where maintenance is required to restore proper operation, the panel shall be removed from test, repaired and replaced on test. All nonoperating time shall be subtracted from the total time, however, each panel shall be tested for a minimum of 200 hours. Each failure shall be recorded by unit serial number, time sequence of operational hours to failure and the station or component where failure occurred. At completion of test, each known uncorrected failure shall be corrected and the maintenance actions recorded. Perform the functional characteristics test specified in 4.6.1 for each panel and record undetected failures or deviations from specified requirements. Any determination as to confirmed or suspected causes of each failure shall be recorded. For purposes of this test, a failure is defined as any random malfunction resulting in inability of any circuit to operate, performance not within the functional characteristics, or any condition that would create a safety hazard for personnel or equipment. The MTBF and 90 percent confidence intervals shall be determined from the accumulated data.

4.7 Inspection comparison. The Government may select panels at any time during the contract production period and subject these panels to the examination and tests specified in 4.5 to determine that the quality of the selected panels is equal to the quality standards established during initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Panels will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction or initial production panels. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of any panel or component, as applicable, to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the panel or component and shall be cause for refusal by the Government to continue acceptance of production panels or components until evidence has been provided by the contractor that corrective action has been taken to

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eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on panels or components produced and accepted under the contract since approval of the initial production panels or components. Any deficiencies found as a result of inspection comparison inspection will be considered prima facie evidence that all panels or components accepted after approval of the initial production panel or components are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purposes of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Examination. Examination shall be performed on samples selected in accordance with the procedures of MIL-STD-105. The AQL shall be 2.5 percent defective. Samples shall be examined for the following defects:

101.	Cable assemblies not coiled, placed, and cushioned as specified.
102.	Technical manuals missing, if required.
103.	Case not closed, secured, and tamperproof seals attached as specified.
104.	Box not as specified for Level A or B.
105.	Pallets not as specified for Level A or B.
106.	Load arrangement not as specified for Level A or B.
107.	Strapping not as specified for Level A or B.
108.	Marking missing, incomplete, incorrect, or illegible for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation (see 6.6). All cable assemblies shall be individually coiled and placed in the compartment for cables. Cushioning of cable assemblies and insertion of the technical manual shall provide a snug fit of items in the case half after securing the compartment cover. The case sections shall be joined together by means of the attached clasps and a minimum of two tamperproof seals attached to the clasps in a manner that the case cannot be opened without destroying the seals.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

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5.2.1 Level A. Each panel, preserved as specified in 5.1, shall be individually packed in the upright position in a close-fitting box conforming to PPP-B-636, V3c. Closure shall be in accordance with the appendix to the box specification. Unless otherwise specified (see 6.2), the packed panels shall be palletized on pallets conforming to NN-P-71, Type IV. Load arrangement shall be two tiers high and shall conform to pallet pattern 90, Appendix D of MIL-STD-147. The packed panels shall be secured to the pallet with a total of four primary and secondary straps. Strapping shall conform to QQ-S-781, Type I or IV, Finish B, size as applicable.

5.2.2 Level B. Each panel, preserved as specified in 5.1, shall be individually packed as specified in 5.2.1 except the box shall be Class Domestic, Grade 200, and strapping shall be Finish A.

5.2.3 Commercial. The panels shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking (see 6.7).

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial packages shall contain the complete item name, National Stock Number, contract number and appropriate address markings. The markings shall be applied with a waterproof material, be clearly legible and provide a definite contrast.

6. NOTES

6.1 Intended use. The panels are intended to be used with a 28 volt DC generator for charging lead acid type batteries in isolated motor pool operations and other areas where charging capabilities using utility electric power are not available.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of panels required for preproduction inspection (see 3.2).
(c) Time frame for initiation of preproduction inspection (see 3.2).
(d) The extent of preproduction inspection to be performed by the Government, when specified (see 3.2).
(e) Number of panels to be submitted for initial production inspection, when specified (see 3.3). Recommend one for group 1 test and a minimum of four for group 2 tests (see 4.5.1.2).

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(f) Spare parts required to support maintainability during reliability test (see 3.13).
(g) Level of packing required (see 5.2).
(h) When palletization is not required for Level A (see 5.2.1) or Level B (see 5.2.2).
(i) Technical publications required (see 6.4).

6.3 Preproduction model. Any changes or deviations of production panels from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish panels conforming to this specification.

6.4 Data requirements. Technical manuals should be furnished as GFE in the contract for overpack and shipment with each panel. If manuals are not furnished GFE, then requirements for technical publications should include operating instructions, safety provisions, illustrated parts list, and FSCM numbers of spare and repair parts. The conventional TM format is recommended.

6.5 Provisioning. Spare and repair parts support should employ the standard inventory established for this item. If new areas of utilization are anticipated, preparations should be made to organize the support necessary for the maintenance levels allocated.

6.6 Preservation (see 5.1). Levels of preservation have not been included as the preservation specified is the minimum acceptable for any level of protection.

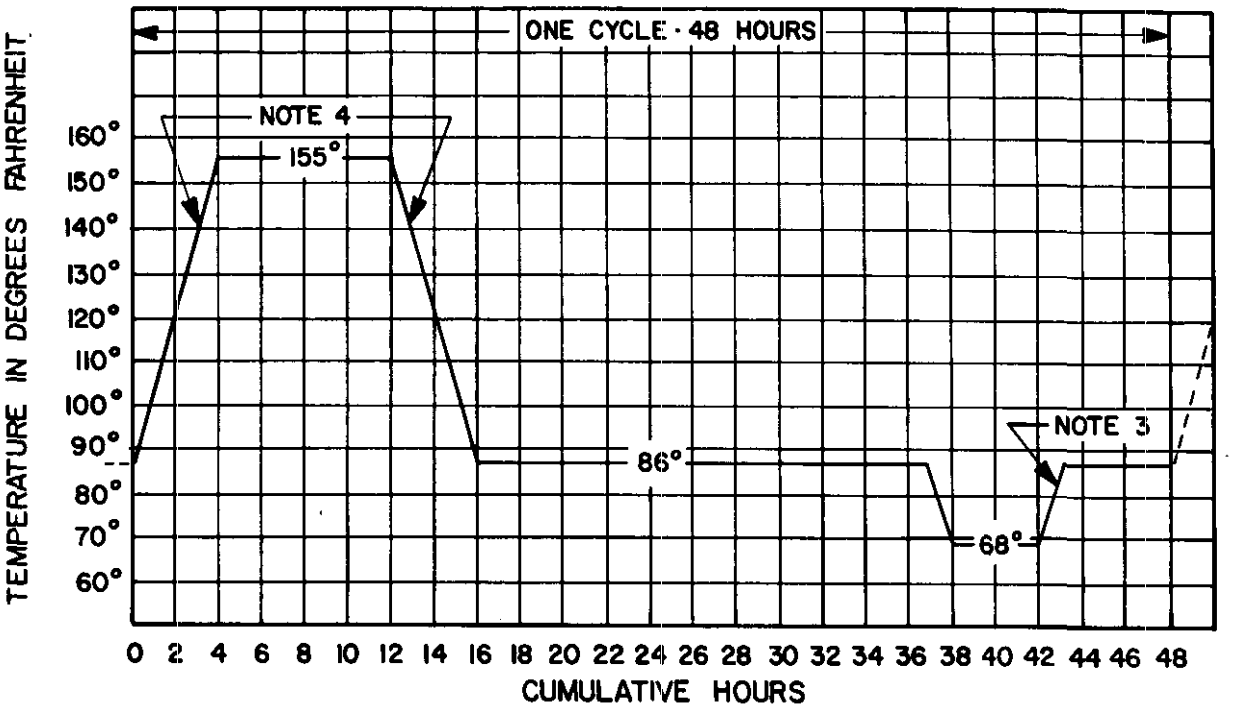
6.7 Marking (see 5.3). For purposes of identification when in storage, the preservation/packing levels shall be marked A/A, A/B, or A/X as applicable.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 6130-A182

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NOTES:

1. THE ACTUAL TEMPERATURE DURING THE CYCLE SHALL BE WITHIN 5° F OF THE TEMPERATURE SHOWN ON THE CHART.
2. RELATIVE HUMIDITY SHALL BE MAINTAINED BETWEEN 90 AND 98% AT ALL TIMES DURING THE CYCLE.
3. THE MEASURED INCREASE IN TEMPERATURE FROM 68 ± 5° F TO 86± 5° F SHALL BE NOT LESS THAN 18° F.
4. RATE OF TEMPERATURE CHANGE BETWEEN 86°F AND 155° F SHALL BE NOT LESS THAN 15°F PER HOUR.

FIGURE 1. Temperature- humidity cycle.

X-2549A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-52457B(ME) Panel, Distribution, Battery Charging

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

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